EchoPulse

Smart Flood Monitoring & Sustainable Water Management (SFM)

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System Overview

An intelligent system based on FMCW Green Radar and AI to predict floods early, reduce energy consumption, and reuse floodwater for agriculture.

Problem statement

Jordan is one of the world's most water-scarce countries, yet it is prone to flash floods that cause significant human and material losses, particularly in areas like Petra, Wadi Mujib, and Zarqa. Current systems consume a lot of energy, provide delayed warnings, and do not utilize floodwater after the danger has passed.

Proposed Solution

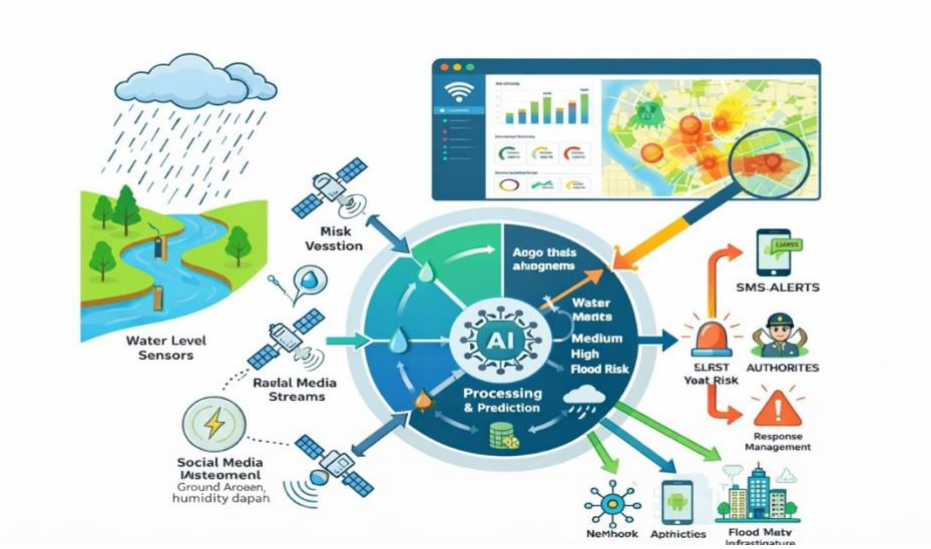
EchoPulse is an intelligent system that uses:
Low-power sensors for continuous monitoring FMCW Radar that activates only when there is danger AI for flood prediction 6–24 hours in advance IoT network for instant connectivity Water quality analysis for water reuse

Key Innovation

- Event-Driven Green Radar : Saves up to 75% more energy compared to conventional systems.
- Al Flood Prediction :Utilizes LSTM + Random Forest for early prediction and risk assessment.
- Smart Water Reuse : Analyzes floodwater quality and reuses it for irrigation.
- Solar Powered System
- Interactive Risk Dashboard: An interactive map displaying the real-time risk level.
- IoT Connectivity: instant alerts to authorities and communities

System Workflow

Sensors → Radar Activation → AI Analysis → Alert System → Water Diversion & Reuse



Impact

Environmental Impact

- 75% less energy
- 40% water reuse for irrigation
- Support for sustainability

Social Impact:

- Protecting lives Early
- warnings for hazardous areas

Economic Impact:

- Reduced infrastructure damage
- Providing water for the agricultural sector

Feasibility & Scalability

The project is based on low-cost components and can be implemented gradually in high-risk areas and then expanded across the Kingdom.

Conclusion

EchoPulse transforms floods from a devastating threat into a sustainable water resource, combining AI, IoT, and green radar to protect lives, preserve energy, and strengthen Jordan's water security.